

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280525

Kind Code

A1

Publication Date

September 04, 2025

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MEMORY DEVICE AND FABRICATING METHOD THEREOF

Abstract

A memory device includes a capacitor contact structure, a capacitor, a bit line structure, and a spacer structure. The capacitor contact structure is disposed on a semiconductor substrate. The capacitor is disposed on the capacitor contact structure. The capacitor extends from a top surface of the capacitor contact structure to a sidewall of the capacitor contact structure. The bit line structure is disposed on the semiconductor substrate and adjacent to the capacitor contact structure. The spacer structure is between the bit line structure and the capacitor contact structure.

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Family ID: 96856678

Appl. No.: 18/593904

Filed: March 02, 2024

Publication Classification

Int. Cl.: H10B12/00 (20230101)

U.S. Cl.:

CPC H10B12/315 (20230201); H10B12/03 (20230201); H10B12/482 (20230201);

Background/Summary

BACKGROUND

Field of Invention

[0001] The present disclosure relates to a memory device and a fabricating method thereof.

Description of Related Art

[0002] Memory cells in the dynamic random access memory (DRAM) have been scaled down continuously to integrate a larger number of the memory cells in a unit area. However, some issues of fabricating the memory cells may arise from the scaling down process. For example, a landing pad is typically formed to connect a capacitor and a capacitor contact structure in the DRAM. However, the landing pad increases the resistance of the capacitor contact structure and deteriorates speed performance of the DRAM.

[0003] Accordingly, how to provide a memory device and a fabricating method thereof to solve the aforementioned problems becomes an important issue to be solved by those in the industry.

SUMMARY

[0004] An aspect of the disclosure is to provide a memory device and a fabricating method thereof that may efficiently solve the aforementioned problems.

[0005] According to an embodiment of the disclosure, a memory device includes a capacitor contact structure, a capacitor, a bit line structure, and a spacer structure. The capacitor contact structure is disposed on a semiconductor substrate. The capacitor is disposed on the capacitor contact structure. The capacitor extends from a top surface of the capacitor contact structure to a sidewall of the capacitor contact structure. The bit line structure is disposed on the semiconductor substrate and adjacent to the capacitor contact structure. The spacer structure is between the bit line structure and the capacitor contact structure.

[0006] In some embodiments of the present disclosure, the spacer structure includes a first spacer along a sidewall of the bit line structure and a second spacer along a sidewall of the capacitor contact structure.

[0007] In some embodiments of the present disclosure, the first spacer and the second spacer are made of different materials.

[0008] In some embodiments of the present disclosure, the first spacer further extends to the sidewall of the capacitor contact structure.

[0009] In some embodiments of the present disclosure, the first spacer is in contact with a top surface of the second spacer.

[0010] In some embodiments of the present disclosure, the capacitor contact structure includes a lower contact plug and an upper contact plug over the lower contact plug. The top surface of the second spacer is lower than a top surface of the lower contact plug.

[0011] In some embodiments of the present disclosure, the spacer structure extends to a top surface of the capacitor contact structure.

[0012] In some embodiments of the present disclosure, a bottom surface of the capacitor is in contact with the spacer structure.

[0013] In some embodiments of the present disclosure, the bit line structure further includes a bit line and a cap layer over the bit line. The capacitor is in contact with the cap layer.

[0014] In some embodiments of the present disclosure, a bottom surface of the capacitor is lower than a top surface of the cap layer.

[0015] In some embodiments of the present disclosure, the bit line structure further includes a bit line and a cap layer over the bit line. The spacer structure is in contact with a top surface of the cap layer.

[0016] According to another embodiment of the disclosure, a fabricating method of a memory device includes forming a bit line structure over a semiconductor substrate. The fabricating method further includes forming a first spacer along the bit line structure. The fabricating method further includes forming a second spacer along the first spacer. The fabricating method further includes forming a capacitor contact structure along the second spacer. The fabricating method further includes etching back the second spacer to form a gap between the first spacer and the capacitor contact structure. The fabricating method further includes forming a third spacer filling the gap and over the first spacer. The fabricating method further includes etching the first spacer and the third

spacer to expose the capacitor contact structure. The fabricating method further includes forming a capacitor electrically connected with the capacitor contact structure.

[0017] In some embodiments of the present disclosure, etching the first spacer and the third spacer is performed to expose a top surface and a sidewall of the capacitor contact structure.

[0018] In some embodiments of the present disclosure, the capacitor is in contact with the top surface and the sidewall of the capacitor contact structure.

[0019] In some embodiments of the present disclosure, etching the first spacer and the third spacer is performed to expose a cap layer of the bit line structure. The capacitor is in contact with the bit line structure.

[0020] In some embodiments of the present disclosure, the capacitor contact structure includes a lower contact plug and an upper contact plug over the lower contact plug. Etching back the second spacer is performed such that a top surface of the etched second spacer is lower than a top surface of the lower contact plug.

[0021] In some embodiments of the present disclosure, the first spacer and the third spacer include a same material.

[0022] In some embodiments of the present disclosure, the third spacer further extends to a top surface of the bit line structure.

[0023] In some embodiments of the present disclosure, the third spacer is in contact with a sidewall of the capacitor.

[0024] In some embodiments of the present disclosure, the third spacer is in contact with a top surface of the second spacer.

[0025] Accordingly, in the memory device of some embodiments of the present disclosure, through the direct and multi-planar connection between the capacitor contact structure and the capacitor, the resistance of the capacitor contact structure may be reduced and the speed performance of the memory device may be improved. In addition, in the fabricating method of the memory device of some embodiments of the present disclosure, the process of forming a landing pad is omitted and the cost of the fabrication may be reduced.

[0026] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0028] FIG. 1 is a circuit diagram of a memory device according to some embodiments of the present disclosure;

[0029] FIG. 2 is a top view of a memory device according to some embodiments of the present disclosure;

[0030] FIG. 3 is a cross-sectional view of a memory device taken along a line 3-3 in FIG. 2 according to some embodiments of the present disclosure; and

[0031] FIG. 4 to FIG. 10 are cross-sectional views of intermediate stages of a fabricating method of a memory device according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0032] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts. However, specific structural and functional details disclosed herein are merely representative for

purposes of describing example embodiments, and thus may be embodied in many alternate forms and should not be construed as limited to only example embodiments set forth herein. Therefore, it should be understood that there is no intent to limit example embodiments to the particular forms disclosed, but on the contrary, example embodiments are to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure.

[0033] Reference is made to FIG. 1. FIG. 1 is a circuit diagram of a memory device **10** according to some embodiments of the present disclosure. As shown in FIG. 1, the memory device **10** (e.g., dynamic random access memory, DRAM) may include a plurality of memory cells MC. A typical DRAM memory cell incorporates a capacitor **180** and a transistor TR. The capacitor **180** temporarily store data based on the charged state of the capacitor **180**. A bit line **122** is electrically connected to a source region of the transistor TR. A word line WL is electrically connected to a gate region of the transistor TR. The capacitor **180** is electrically connected to the corresponding transistor by a capacitor contact structure (not shown in FIG. 1).

[0034] Reference is made to FIG. 2. FIG. 2 is a top view of the memory device **10** according to some embodiments of the present disclosure. It should be noted that FIG. 2 only illustrates some of the components of the memory device **10**, and other components such as the capacitor **180** is omitted for simplicity. As shown in FIG. 2, active areas AA, bit lines **122**, and word lines WL are formed. To be more specific, the bit lines **122** extend along a first direction D1 and the word lines WL extends along a second direction D2 perpendicular to the first direction D1.

[0035] Reference is made to FIG. 3. FIG. 3 is a cross-sectional view of the memory device **10** taken along a line 3-3 in FIG. 2 according to some embodiments of the present disclosure. As shown in FIG. 3, the memory device **10** includes a semiconductor substrate **100**, a bit line structure **120**, a capacitor contact structure **140**, a spacer structure **160**, and a capacitor **180**.

[0036] In some embodiments, the word lines WL are embedded in the active area AA of a substrate. A gate dielectric layer (not shown) may line the surface of each word line WL. Source/drain regions are in the active area AA and on opposite sides of each word line WL. The source/drain regions may include opposite conductivity type than the active area AA. Accordingly, the word line WL, the gate dielectric layer, the source/drain regions, and the active area AA may collectively serve as the transistor TR of the memory cell MC, in which the word line WL and the gate dielectric layer may serve as a gate structure of the transistor TR.

[0037] As shown in FIG. 3, the bit line structure **120** and the capacitor contact structure **140** are disposed on the semiconductor substrate **100**. The bit line structure **120** is adjacent to the capacitor contact structure **140**. The spacer structure **160** is disposed between the bit line structure **120** and the capacitor contact structure **140**. The capacitor **180** is disposed on the capacitor contact structure **140**. The capacitor **180** is electrically connected to its corresponding transistor through the capacitor contact structure **140**. In the present disclosure, the capacitor **180** extends from a top surface of the capacitor contact structure **140** to a sidewall of the capacitor contact structure **140**. As such, the connection between the capacitor contact structure **140** and the capacitor **180** is direct and multi-planar. Thus, a typical landing pad may be omitted, and the contact resistance between the capacitor contact structure **140** and the capacitor **180** is thereby reduced.

[0038] To be more specific, as shown in FIG. 3, the bit line structure **120** includes a bit line **122**, a cap layer **124**, and a bit line contact **126**. The bit line **122** is disposed on the semiconductor substrate **100** through the bit line contact **126**. The cap layer **124** is disposed over the bit line **122**. In some embodiments, the bit line **122** includes conductive materials, such as tungsten. The cap layer **124** includes dielectric materials, such as silicon nitride (SIN). The bit line contact **126** may include conductive materials, such as polysilicon.

[0039] As shown in FIG. 3, the capacitor contact structure **140** includes a lower contact plug **142** and an upper contact plug **144** over the lower contact plug **142**. In some embodiments, the lower contact plug **142** includes poly-crystalline silicon (polysilicon), and the upper contact plug **144** includes conductive materials, such as tungsten.

[0040] As shown in FIG. 3, the spacer structure **160** includes a first spacer **162** along a sidewall of the bit line structure **120**. To be more specific, the first spacer **162** is disposed along sidewalls of the bit line **122**, the cap layer **124**, and the bit line contact **126**. In some embodiments, the first spacer **162** further extends over and contacts a top surface of the cap layer **124**. In some embodiments, the first spacer **162** may include a spacer material **162-1** and a spacer material **162-2**. [0041] The spacer structure **160** further includes a second spacer **164** along a sidewall of the capacitor contact structure **140**, as shown in FIG. 3. To be more specific, the second spacer **164** is disposed along a sidewall of the lower contact plug **142**, surrounding and in contact with the lower contact plug **142**.

[0042] In some embodiments, the first spacer **162** extends to a sidewall of the capacitor contact structure **140**. In greater detail, the first spacer **162** extends to sidewalls of the lower contact plug **142** and the upper contact plug **144**. The first spacer **162** then extends and covers a top surface **164a** of the second spacer **164**.

[0043] In some embodiments, the first spacer **162** further extends over and contacts the top surface of the capacitor contact structure **140** (e.g., a top surface **144a** of the upper contact plug **144**).

[0044] The spacer structure **160** includes dielectric materials. In some embodiments, the first spacer **162** and the second spacer **164** are made of different materials. For example, the first spacer **162** includes silicon nitride, and the second spacer **164** includes silicon oxide.

[0045] As shown in FIG. 3, the capacitor **180** extends from the top surface **144a** to a sidewall **144b** of the upper contact plug **144**. As such, the connection between the capacitor contact structure **140** and the capacitor **180** is direct at the top surface **144a** and the sidewall **144b**.

[0046] In some embodiments, a bottom surface of the capacitor **180** is in direct contact with the first spacer **162** of the spacer structure **160**. In addition, a sidewall of the capacitor **180** may also be in contact with the first spacer **162**. In some embodiments, the capacitor **180** may also be in contact with the cap layer **124** of the bit line structure **120**. To be more specific, the bottom surface and the sidewall of the capacitor **180** may also be in contact with the cap layer **124**.

[0047] Reference is made to FIG. 4 to FIG. 10. FIG. 4 to FIG. 10 are cross-sectional views of intermediate stages of a fabricating method of the memory device **10** according to some embodiments of the present disclosure. Operations of the fabricating method will be described accompanied with the drawings.

[0048] Reference is made to FIG. 4. The fabricating method starts with forming a bit line structure **120** on a semiconductor substrate **100**. The semiconductor substrate **100** may include structures such as word lines as aforementioned.

[0049] Next, the fabricating method includes forming a spacer material **162-1** along the bit line structure **120** and then forming a spacer material **164** along the spacer material **162-1**. In some embodiments, the spacer material **162-1** surrounds the bit line structure **120** and contacts sidewalls of a bit line **122**, a cap layer **124**, and a bit line contact **126** of the bit line structure **120**.

[0050] Reference is made to FIG. 5 to FIG. 7. Subsequently, the fabricating method includes forming a capacitor contact structure **140** along the spacer material **164**. To be more specific, the capacitor contact structure **140** includes a lower contact plug **142** and an upper contact plug **144** over the lower contact plug **142**. As shown in FIG. 5, the lower contact plug **142** is firstly disposed along the second spacer **164**. Then, as shown in FIG. 6, the lower contact plug **142** is etched back such that a top surface **142a** of the lower contact plug **142** is lower than a top surface of the spacer material **164**. Next, the upper contact plug **144** is formed over the etched lower contact plug **142** and a top surface **144a** of the upper contact plug **144** is substantially level with the top surface of the spacer material **164**, as shown in FIG. 7.

[0051] Reference is made to FIG. 8. The fabricating method includes etching back the spacer material **164** to form a gap **G** between the spacer material **162-1** and the capacitor contact structure **140**, as shown in FIG. 8. In some embodiments, etching back the spacer material **164** is performed such that a top surface **164a** of the etched spacer material **164** is lower than the top surface **142a** of

the lower contact plug **142** by a height H. In other words, after the etching process, the upper contact plug **144** and a portion of the lower contact plug **142** protrude from the top surface **164a** of the etched spacer material **164**. The etched spacer material **164** is also referred to as the second spacer **164**.

[0052] Reference is made to FIG. **9**. The fabricating method includes forming a spacer material **162-2** filling the gap G and over the spacer material **162-1**, as shown in FIG. **9**. In some embodiments, the spacer material **162-2** further extends to a top surface of the bit line structure **120** (e.g., a top surface of the cap layer **124**). In some embodiments, the spacer material **162-2** is in contact with the top surface **164a** of the second spacer **164**. In some embodiments, the spacer material **162-2** is in contact with a sidewall of the capacitor contact structure **140** (e.g., the sidewall **144b** of the upper contact plug **144**). In some embodiments, the spacer material **162-1** and the spacer material **162-2** include a same material, such as silicon nitride. In other embodiments, the spacer material **162-1** and the spacer material **162-2** include different dielectric materials. The spacer material **162-1** and the spacer material **162-2** are collectively referred to as the first spacer **162**. The first spacer **162** and the second spacer **164** are collectively referred to as the spacer structure **160**.

[0053] Reference is made to FIG. **10**. The fabricating method includes etching the spacer material **162-1** and the spacer material **162-2** to expose the capacitor contact structure **140**. To be more specific, after the etching process, a top surface and a sidewall of the capacitor contact structure **140** (e.g., the top surface **144a** and the sidewall **144b** of the upper contact plug **144**) are exposed through the resultant trench T, as shown in FIG. **10**. In some embodiments, as shown in FIG. **10**, a portion of the cap layer of the bit line structure **120** is etched away after the etching process and the trench T exposes the cap layer **124**.

[0054] Reference is made back to FIG. **3**. The fabricating method includes forming a capacitor **180** electrically connected with the capacitor contact structure **140**. To be more specific, the capacitor **180** is formed in the trench T. As shown in FIG. **3**, the capacitor **180** is in contact with the exposed top surface and sidewall of the capacitor contact structure **140** (e.g., the top surface **144a** and the sidewall **144b** of the upper contact plug **144**). In addition, the etched spacer material **162-2** is in contact with a sidewall of the capacitor **180**. In embodiments where the cap layer **124** is exposed through the trench T, the capacitor **180** formed in the trench T is in contact with the bit line structure **120**, or more particularly, with the cap layer **124**.

[0055] As shown in FIG. **3**, the capacitor **180** may include a bottom electrode, a dielectric layer, and a top electrode. During the fabrication of the capacitor **180**, the bottom electrode and the dielectric layer may be conformally deposited in the trench T. As such, the bottom electrode and the dielectric layer of the capacitor **180** have a stepped profile to blanket-cover and contact the capacitor contact structure **140**, as shown in FIG. **3**.

[0056] Accordingly, in the memory device of some embodiments of the present disclosure, through the direct and multi-planar connection between the capacitor contact structure and the capacitor, the resistance of the capacitor contact structure may be reduced and the speed performance of the memory device may be improved. In addition, in the fabricating method of the memory device of some embodiments of the present disclosure, the process of forming a landing pad is omitted and the cost of the fabrication may be reduced.

[0057] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0058] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure covers modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A memory device, comprising: a capacitor contact structure disposed on a semiconductor substrate; a capacitor disposed on the capacitor contact structure, wherein the capacitor extends from a top surface of the capacitor contact structure to a sidewall of the capacitor contact structure; a bit line structure disposed on the semiconductor substrate and adjacent to the capacitor contact structure; and a spacer structure between the bit line structure and the capacitor contact structure.
2. The memory device of claim 1, wherein the spacer structure comprises a first spacer along a sidewall of the bit line structure and a second spacer along a sidewall of the capacitor contact structure.
3. The memory device of claim 2, wherein the first spacer and the second spacer are made of different materials.
4. The memory device of claim 2, wherein the first spacer further extends to the sidewall of the capacitor contact structure.
5. The memory device of claim 4, wherein the first spacer is in contact with a top surface of the second spacer.
6. The memory device of claim 5, wherein the capacitor contact structure comprises a lower contact plug and an upper contact plug over the lower contact plug, and wherein the top surface of the second spacer is lower than a top surface of the lower contact plug.
7. The memory device of claim 1, wherein the spacer structure extends to a top surface of the capacitor contact structure.
8. The memory device of claim 1, wherein a bottom surface of the capacitor is in contact with the spacer structure.
9. The memory device of claim 1, wherein the bit line structure further comprises a bit line and a cap layer over the bit line, and the capacitor is in contact with the cap layer.
10. The memory device of claim 9, wherein a bottom surface of the capacitor is lower than a top surface of the cap layer.
11. The memory device of claim 1, wherein the bit line structure further comprises a bit line and a cap layer over the bit line, and the spacer structure is in contact with a top surface of the cap layer.
12. A fabricating method of a memory device, comprising: forming a bit line structure over a semiconductor substrate; forming a first spacer along the bit line structure; forming a second spacer along the first spacer; forming a capacitor contact structure along the second spacer; etching back the second spacer to form a gap between the first spacer and the capacitor contact structure; forming a third spacer filling the gap and over the first spacer; etching the first spacer and the third spacer to expose the capacitor contact structure; and forming a capacitor electrically connected with the capacitor contact structure.
13. The fabricating method of claim 12, wherein etching the first spacer and the third spacer is performed to expose a top surface and a sidewall of the capacitor contact structure.
14. The fabricating method of claim 13, wherein the capacitor is in contact with the top surface and the sidewall of the capacitor contact structure.
15. The fabricating method of claim 12, wherein etching the first spacer and the third spacer is performed to expose a cap layer of the bit line structure, and the capacitor is in contact with the bit line structure.
16. The fabricating method of claim 12, wherein the capacitor contact structure comprises a lower contact plug and an upper contact plug over the lower contact plug, and wherein etching back the second spacer is performed such that a top surface of the etched second spacer is lower than a top surface of the lower contact plug.
17. The fabricating method of claim 12, wherein the first spacer and the third spacer comprise a same material.

- 18.** The fabricating method of claim 12, wherein the third spacer further extends to a top surface of the bit line structure.
- 19.** The fabricating method of claim 12, wherein the third spacer is in contact with a sidewall of the capacitor.
- 20.** The fabricating method of claim 12, wherein the third spacer is in contact with a top surface of the second spacer.
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